MACROEVOLUTION

Module map

- Module 14: Macroevolution
- Connected lab: lab-14_macroevolution.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 14

Macroevolution

1. Species concepts
2. Reproductive isolation
3. Prezygotic barriers
4. Postzygotic barriers
5. Speciation patterns and phylogeny

Lab evidence

lab-14_macroevolution.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

MACROEVOLUTION

Learning objectives

- State the biological species concept and its limits.
- Explain reproductive isolation as reduced gene flow.
- Classify examples as prezygotic or postzygotic barriers.
- Compare allopatric and sympatric speciation.
- Interpret speciation as population-level divergence over time.

OBJECTIVE 1
State the biological species concept and its limits.

OBJECTIVE 2
Explain reproductive isolation as reduced gene flow.

OBJECTIVE 3
Classify examples as prezygotic or postzygotic barriers.

OBJECTIVE 4
Compare allopatric and sympatric speciation.

OBJECTIVE 5
Interpret speciation as population-level divergence over time.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

MACROEVOLUTION

Topic sequence

- Species concepts: Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
- Reproductive isolation: Reproductive isolation reduces gene flow between populations.
- Prezygotic barriers: Prezygotic barriers prevent mating or fertilization before a zygote forms.
- Postzygotic barriers: Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.
- Speciation patterns and phylogeny: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

01

Species concepts

Species concepts define lineages using reproductive, morphological, or evolutionary criteria.

02

Reproductive isolation

Reproductive isolation reduces gene flow between populations.

03

Prezygotic barriers

Prezygotic barriers prevent mating or fertilization before a zygote forms.

04

Postzygotic barriers

Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.

05

Speciation patterns and phylogeny

Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

MACROEVOLUTION

Module 14: Macroevolution Evidence Map

- Module focus: Macroevolution
- Central claim: Macroevolution explains speciation as reduced gene flow, reproductive isolation, and phylogenetic divergence.
- Key nodes to connect: Species concepts, Reproductive isolation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-14_macroevolution.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

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Module 14: Macroevolution Reasoning Sequence

- Module focus: Macroevolution
- Trace the model: Species concepts -> Reproductive isolation -> Prezygotic barriers -> Postzygotic barriers
- Inputs become outputs: Species concepts, Reproductive isolation -> State the biological species concept and its limits., Explain reproductive isolation as reduced gene flow.
- Feedback check: lab-14_macroevolution.md evidence checks whether students can explain prezygotic barriers.
- Lab connection: lab-14_macroevolution.md gives students a process to observe or test.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

MACROEVOLUTION

Terms as evidence handles

- Species: Lineage or group identified by a species concept.
- Biological species concept: Species defined by interbreeding and reproductive isolation.
- Reproductive isolation: Barriers that reduce gene flow between groups.
- Prezygotic barrier: Barrier before fertilization.
- Postzygotic barrier: Barrier after fertilization.
- Allopatric speciation: Speciation with geographic separation.

SPECIES
Lineage or group identified by a species concept.

BIOLOGICAL SPECIES CONCEPT
Species defined by interbreeding and reproductive isolation.

REPRODUCTIVE ISOLATION
Barriers that reduce gene flow between groups.

PREZYGOTIC BARRIER
Barrier before fertilization.

POSTZYGOTIC BARRIER
Barrier after fertilization.

ALLOPATRIC SPECIATION
Speciation with geographic separation.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

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Lab connection

- Lab file: lab-14_macroevolution.md
- Lab evidence should test or illustrate: Prezygotic barriers
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Species concepts

EVIDENCE
lab-14_macroevolution.md

CLAIM
State the biological species concept and its limits.

Teaching note: Make the lab connection explicit before students encounter the activity.

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Contrast check

- Do not stop at: Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
- Stronger explanation: Speciation can occur allopatrically or sympatrically and is represented in phylogenies.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Species concepts define lineages using reproductive, morphological, or evolutionary criteria.

DEEPER
Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

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Module 14: Macroevolution Retrieval and Lab Check

- Module focus: Macroevolution
- Retrieval prompt: What does the biological species concept emphasize?
- Answer check: State the biological species concept and its limits.
- Required terms: Species, Biological species concept, Reproductive isolation
- Lab connection: lab-14_macroevolution.md; lab-14_macroevolution.md supplies evidence for isolation, lineage splitting, and phylogenetic pattern evidence.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

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Practice quiz bridge

- 1. Under the biological species concept, two groups count as different species when they:
- 2. Two related species breed at different times of year and so never mate. This is an example of a:
- 3. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
- 4. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

QUESTION 1

Key: C

Under the biological species concept, two groups count as different species when they:

QUESTION 2

Key: A

Two related species breed at different times of year and so never mate. This is an example of a:

QUESTION 3

Key: D

A mountain range splits one population into two groups that slowly diverge into separate species. This is:

QUESTION 4

Key: B

A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

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Synthesis and exit ticket

- Exit claim: explain species concepts using evidence from lab-14_macroevolution.md.
- Must include: Species, Biological species concept, Reproductive isolation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Species concepts

EVIDENCE
lab-14_macroevolution.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.